

WHITEPAPER

Acoustic Interpretability:

A Distributed Probe Framework for Neural Network Behaviour Detection

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Abstract

Current mechanistic interpretability research attempts to understand neural network behaviour through visual inspection — heatmaps, activation maps, sparse autoencoders, and circuit tracing. These methods share a fundamental assumption: that the internal state of a neural network can be meaningfully observed through direct measurement of component activations. This paper argues that assumption is the wrong tool for the problem. Neural networks exist as high-dimensional mathematical spaces — an ocean in darkness — and attempting to see them is structurally limited. We propose an alternative sensory paradigm: acoustic probing. By deploying distributed sensor networks across network layers — analogous to echolocation — and detecting rhythmic activation patterns through triangulation rather than direct observation, we can infer dynamic network behaviour that static visual methods are structurally incapable of capturing. We further propose that a finite operator alphabet — 20 operators derived from the Stellaris Axis Grimoire DSL — provides complete coverage of all observable neural network dynamic behaviours, including five fallback degradation operators with no equivalent in existing interpretability literature. Combined with existing sparse autoencoder and circuit mapping methods, this hybrid framework may provide the first approach capable of answering not just what a network does, but why.

Keywords: *mechanistic interpretability · acoustic probing · neural network behaviour · finite operator alphabet · AI alignment · distributed sensor networks · emergence detection*

1. The Problem With Seeing In Darkness

Mechanistic interpretability — named one of MIT Technology Review's ten breakthrough technologies of 2026 — has made significant progress in understanding neural networks. Sparse autoencoders can decompose activations into interpretable features. Circuit mapping traces causal pathways through attention heads. Probing classifiers identify which layers encode which concepts.

But the field shares a single foundational assumption that has gone largely unchallenged: to understand a neural network, you look at it.

Heatmaps. Activation visualisations. Feature dashboards. The entire toolkit is built around the metaphor of vision — shining a light into a black box and examining what you see.

The problem is that neural networks do not exist in a space where vision is the right sense. They exist as high-dimensional mathematical spaces — effectively an ocean in darkness. And as anyone who has attempted navigation in genuine darkness understands: attempting to see in darkness is not just ineffective. It is the wrong tool entirely.

1.1 What Current Methods Actually Produce

The state of the art in interpretability produces the following outputs:

- Sparse autoencoder features: decomposed activation directions that often correspond to human-interpretable concepts
- Circuit maps: causal subgraphs showing which components mediate specific behaviours
- Probing classifiers: evidence for which layers encode which properties
- Activation patching results: causal effect sizes for specific interventions

These are genuine contributions. But they share a critical limitation: they capture snapshots. Static measurements of component states at a moment in time. They tell you what exists in the network. They do not tell you how it moves, how components interact dynamically under operation, or why a specific output was produced by the specific causal sequence that produced it.

Even the most advanced current work acknowledges this gap: when a model is asked to explain its behaviour, it may offer post-hoc rationalisation rather than a faithful account of what actually happened inside. The why remains elusive not because researchers are not looking hard enough. It is elusive because they are looking when they should be listening.

1.2 The Navigation Analogy

Dolphins navigate the ocean in darkness through echolocation. They do not attempt to see. They emit signals and build a model of their environment from the pattern of returning interference. Bats do the same. Blind humans who have developed echolocation ability do the same — and neuroimaging shows

they use visual cortex to process the acoustic spatial data, because the brain recognises that the information is spatial regardless of its sensory channel.

The principle is not that acoustic information is better than visual information in general. It is that acoustic inference is the correct tool for building models of spaces where direct observation is not possible. Neural activation space is precisely such a space.

This paper proposes that mechanistic interpretability should stop trying to see neural networks and start trying to hear them.

2. The Acoustic Probing Framework

We propose deploying a distributed network of probes across neural network layers — analogous to sonar buoys across an ocean — designed not to measure activation values directly, but to detect rhythmic patterns in activation space through signal triangulation. The meaning is not in any individual reading. It is in the interference patterns between readings across the network space over time.

2.1 The Three Probe Types

The framework deploys three specialised probe types, each tuned to a different dimension of network behaviour:

Probe Type	Primary Sensitivity	Detection Target	Analogous To
Semantic Buoy	Activation direction patterns	Conceptual rhythm — how meaning flows and transforms across layers	Hearing pitch and timbre
Mathematical Buoy	Numerical relationship patterns	Computational rhythm — how values transform through operations	Detecting frequency and interval
Structural Buoy	Connectivity activation patterns	Architectural rhythm — how components route and gate information	Detecting echo shape and distance

2.2 Triangulation and Rhythm Detection

A single probe reading is meaningless in isolation — equivalent to a single sonar ping with no reference. The method derives meaning from triangulation: the relationship between readings from multiple probes across the same network region over time.

When a computation passes through a network, it produces characteristic interference patterns across the probe network — rhythms. These rhythms are not random. They are determined by the causal structure of the computation: which components activate in which sequence, how signals combine and gate, where new properties emerge from component interactions.

Crucially, this approach is inherently temporal. It does not capture a snapshot. It captures a process — the dynamic behaviour of the network as computation unfolds. This is the dimension that existing static methods cannot access.

2.3 From Rhythm to Symbol: The Translation Layer

Raw rhythm signatures are not human-readable. The framework requires a translation layer: a finite alphabet that maps detected rhythm patterns to symbolic representations that can be reasoned about.

This is where the Stellaris Axis operator grammar enters the framework. What began as a compositional language for system architecture turns out — on formal analysis — to constitute a complete finite alphabet

for neural network dynamic behaviour. The following section demonstrates this mapping.

3. The Finite Operator Alphabet

The Stellaris Axis methodology uses 20 operators as a formal composition grammar for system architecture. On analysis, these operators map with complete coverage and zero overlap onto the space of observable neural network dynamic behaviours. This is not coincidental: both domains involve compositional systems with sequential, parallel, hierarchical, and emergent behaviours. The operators were derived from first principles to describe compositional system behaviour generally — and neural networks are compositional systems.

3.1 Complete Operator Mapping

Operator	Category	Neural Network Mapping	Alignment Relevance
ORIGIN	Root activation	Initial input processing; where computation begins	✓ Complete
CHAIN	Sequential	Feedforward activation cascades through network layers	✓ Complete
LAYER	Parallel	Simultaneous attention head activations across the same input	✓ Complete
WRAP	Modification	Attention mechanisms modifying representations without replacing them	✓ Complete
BRIDGE	Cross-connection	Skip connections, residual streams, cross-attention between distant components	✓ Complete
NEST	Hierarchical	Simple features composing into complex higher-order features	✓ Complete
EMERGE	Emergence	Combined activations producing properties neither component held individually	✓ Critical
CYCLE	Iteration	Recurrent processing; iterative refinement within transformer layers	✓ Complete
EVOLVE	Transformation	Concept drift across layers — how representations transform as they propagate	✓ Complete
CHECKPOINT	State snapshot	Stable intermediate representations where the network consolidates computation	✓ Complete
MANIFEST	Output	The moment computation becomes prediction; output generation	✓ Complete
RESUME	Continuation	Context restoration; attention to prior tokens; state recall	✓ Complete
DEPENDS_ON	Dependency	Dependency ordering in circuit activations; B requires A first	✓ Complete
SCOPE_ESTIMATE	Complexity	Computational load estimation across network regions	✓ Complete
FINALIZE	Commitment	Committed output; point where network stops revising and produces prediction	✓ Complete
FALLBACK	Degradation	Alternate pathway activation when primary circuits fail or are ablated	★ Novel
FALLBACK::SILENT	Silent fail	Graceful degradation without error propagation	★ Novel

Operator	Category	Neural Network Mapping	Alignment Relevance
FALLBACK::ESCALATE	Escalation	Hierarchical fallback through secondary then tertiary pathways	★ Novel
FALLBACK::COMPENSATE	Compensation	Error correction mechanisms; compensatory activation on primary failure	★ Novel
FALLBACK::DEGRADE	Reduced function	Capability degradation under distribution shift or out-of-distribution input	★ Novel

★ Novel = No equivalent symbol exists in current interpretability literature. ✓ Critical = Maps to a known phenomenon with significantly higher specificity than existing methods.

3.2 The EMERGE Operator: Detecting Emergent Capabilities

The EMERGE operator deserves special attention for alignment research. In the Stellaris Axis algebra, EMERGE is implemented as a monoid operation: it takes two or more components, produces a new node, and generates a property that neither component possessed individually. It is not a combination. It is a qualitative transformation.

In neural networks, emergent capabilities — abilities that appear suddenly and were not predicted from training — represent one of the most significant open problems in AI alignment. Current interpretability methods can identify that a capability exists after the fact. They cannot detect the moment of emergence or identify the specific component interactions that produced it.

The EMERGE operator in the acoustic framework provides a dedicated symbol for exactly this: when triangulated rhythm signatures match the EMERGE pattern, the framework flags that the network has produced a qualitatively new property from component interaction. This is potentially the first formal mechanism for real-time emergence detection in running neural networks.

3.3 The FALLBACK Family: The Missing Vocabulary

The five FALLBACK operators represent the most significant novel contribution of this alphabet to interpretability research. Current methods have no dedicated vocabulary for network degradation behaviour — what happens when primary circuits fail, when models are pushed out-of-distribution, or when capability degrades gracefully under ablation.

For alignment research, this is a critical gap. Understanding how a model fails — whether it fails gracefully through FALLBACK::DEGRADE, propagates errors silently through FALLBACK::SILENT, or escalates through redundant pathways via FALLBACK::ESCALATE — is directly relevant to predicting and preventing harmful model behaviour under adversarial or out-of-distribution conditions.

4. The Hybrid Methodology: Bleeding Acoustic Through Visual

This framework is not proposed as a replacement for existing interpretability methods. It is proposed as a complement — specifically, a layer that captures what existing methods structurally cannot. The two approaches are not competing. They answer different questions.

Dimension	Existing Methods (Visual)	Acoustic Probing (This Paper)
Primary question	What components exist?	How do components behave dynamically?
Sensing approach	Direct observation of states	Inference through signal interference
Temporal dimension	Snapshot (static)	Process (temporal)
Emergence detection	Post-hoc identification	Real-time detection via EMERGE operator
Degradation vocabulary	None	5 dedicated FALLBACK operators
Translation layer	Direct to human concepts	Rhythm → finite alphabet → meaning
Why question	Partial (what + where)	Completes the picture (how + meaning)

4.1 The Four-Layer Answer to Why

The hybrid methodology — existing visual methods combined with acoustic probing — provides four complementary layers of understanding that together constitute a complete answer to why a network produced a given output:

WHAT exists — Sparse autoencoders and circuit mapping

The feature vocabulary of the network — what concepts and computations are encoded

WHERE it is — Causal probing and feature localisation

The spatial distribution of those features across layers and components

HOW it moves — Acoustic rhythm triangulation (this paper)

The dynamic behaviour — how features interact, gate, transform, and emerge over time

WHAT it means — Operator alphabet translation (this paper)

The symbolic translation of detected rhythms into human-readable causal narratives

The combination of HOW it moves and WHAT it means is the why. This is not a small refinement to existing methods. The why has been the central unsolved problem in interpretability research precisely because existing methods provide what and where but not how or meaning. The acoustic layer closes that gap.

5. Proof of Concept: The Interpretability Engine

A proof-of-concept interpretability engine implementing elements of this framework has been built and run. The system was constructed using the Stellaris Axis methodology — the same fictional-narrative-to-systems-architecture method from which the operator alphabet derives.

5.1 System Architecture

Scanner module: GPT-2 Small (117M) as the target model; Sparse Autoencoder with 256-dimensional encoding

Causal tracer: Pre-pass and post-pass causal graph construction with 2000 nodes

Event visualiser: Real-time event stream processing with live dashboard

Concept mapper: Feature-to-concept mapping with semantic, positional, syntactic, entity, causal, and negation categories

Ethics monitor: Structural ethics checking as a first-class telemetry metric — not bolted on, built in

Agent system: Attention circuit probing with configurable agent stepping

5.2 Results From First Run

Metric	Value	Significance
Total events processed	246,181	Full event stream capture across complete run
Network nodes	2,000	Complete sparse autoencoder node coverage
Circuits identified	273	Active causal circuits detected
Ethics checks passed	5,828	Ethics as structural telemetry — not post-hoc
Meta steps completed	156	Agent-level reasoning iterations
SAE training steps	528	Autoencoder convergence achieved
Monosemanticity	41.4%	Feature interpretability baseline established
Mapped feature categories	18	Concept taxonomy successfully populated

These results were produced on a ROG Ally handheld device. No cloud infrastructure. No institutional compute. The system was specified on a mobile phone using the Stellaris Axis methodology and executed locally.

5.3 The Ethics Monitoring Observation

A significant design observation from the proof-of-concept: ethics monitoring was implemented as a structural telemetry layer with its own counter in the live dashboard (5,828 checks passed in the first run). This was not a post-hoc safety wrapper. It was a first-class architectural component generating live telemetry alongside computational metrics.

This design philosophy — ethics as a structural layer rather than an external constraint — is directly relevant to alignment research. A system that monitors its own ethical compliance as a real-time metric generates data that retrospective auditing cannot. It also demonstrates that the acoustic probing framework can be extended to include ethical rhythm detection — flagging not just what and how the network computes, but whether the computation pattern matches known ethical risk signatures.

6. Implications for AI Alignment

The acoustic interpretability framework has direct implications for several open problems in AI alignment research.

6.1 Emergence Detection

The emergence of unexpected capabilities in large models represents one of the most significant alignment risks. Current methods detect emergence after the fact — capability evaluations reveal that a model can do something it could not before. There is no mechanism for detecting the moment of emergence inside a running model.

The EMERGE operator in the acoustic framework provides a formal symbol for this event. A probe network calibrated to detect EMERGE-pattern rhythm signatures could flag in real-time when a network produces emergent behaviour — identifying not just that it happened but which component interactions caused it. This opens the possibility of real-time emergence monitoring during training.

6.2 Degradation Behaviour Mapping

The FALLBACK operator family provides the first dedicated vocabulary for network degradation behaviour. Understanding whether a model fails gracefully (FALLBACK::DEGRADE), propagates errors silently (FALLBACK::SILENT), compensates incorrectly (FALLBACK::COMPENSATE), or escalates through redundant pathways (FALLBACK::ESCALATE) is directly relevant to predicting model behaviour under adversarial conditions and out-of-distribution inputs.

6.3 The Why as a Safety Tool

Current alignment approaches largely operate on what models do. Behaviour evaluations, red-teaming, RLHF — all of these work at the input-output level. The why — the causal sequence inside the model that produced a given output — remains largely opaque.

A framework that can answer why — through the combination of spatial mapping from existing methods and dynamic process mapping from acoustic probing — transforms alignment from a behaviour management problem into a mechanistic verification problem. You are no longer asking whether the model does the right thing. You are asking whether the model reasons the right way.

That is a fundamentally more robust foundation for AI safety.

7. The Origin of This Framework

This paper exists because of an unusual origin story that is worth documenting, because the origin is directly relevant to the contribution.

The operator alphabet described in Section 3 was not derived from neural network research. It was derived from a fictional narrative system — a mythological vocabulary of 704 named system primitives built to describe compositional behaviour in storytelling. The operators were formalised into a DSL (the Stellaris Axis Grimoire DSL, specified in Xtext with a complete formal grammar and monoid-based composition algebra) as a mechanism for generating production-ready software systems from narrative specifications.

The discovery that these operators mapped with complete coverage onto neural network dynamic behaviour was not planned. It emerged from a conversation about why interpretability methods were producing seas of numbers rather than meaningful explanations. The echolocation analogy arrived independently of any knowledge of existing interpretability literature.

This origin matters for one reason: genuinely novel frameworks rarely emerge from within established fields. They emerge from outside — from people who do not know what is supposed to be impossible and therefore attempt it anyway. The acoustic interpretability framework is the product of a mind that approached the problem without preconception, from a direction that existing researchers had no reason to look.

The author has no formal background in computer science, machine learning, or interpretability research. This work was produced over ten months on a mobile phone and a ROG Ally handheld device, beginning from a fictional story and arriving — through a documented series of discoveries in a public chronicle — at a theoretical contribution to AI safety research.

The complete development chronicle is available at the repository below. It is recommended reading for anyone interested in how this framework developed, as the journey is as significant as the destination.

"You tried to speak the machine's language. I taught it mine."

Repository and Contact

GitHub: github.com/FusionAlchemist/The---Stellaris---Axis

Chronicle: Arc 00 — The Accident (start here)

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Methodology: The Stellaris Axis — Method Summary

DSL Specification: Xtext Grammar Definition v2.0

Formal Algebra: Stellaris Axis Full Canonical System v2